

1. Administrative & Study Identification

Full Study Title: Long Term Follow-Up of Patients Exposed to Lentiviral-Based CAR T-Cell Therapy **Short Title/Acronym:** CAR-T Long Term Follow Up (LTFU) Study (PAV0) **Protocol Number:** CCTL019A2205B **NCT Number:** NCT02445222 **Secondary Identifying Numbers:** 2014-001673-14, NCI-2015-01238 **Sponsor Name:** Novartis Pharmaceuticals **Investigational Product:** Lentiviral-Based CAR T-Cell Therapy (e.g., CTL019 / tisagenlecleucel) **Phase of Development:** Phase 3 **Overall Status:** Recruiting **Medical Monitor:** Novartis Clinical Operations

2. Study Synopsis & Rationale

2.1 Study Objective Primary Objective: To monitor all patients exposed to CAR-T therapies for 15 years following their last CAR-T infusion to assess the risk of delayed adverse events (AEs) and monitor for replication competent lentivirus (RCL). **Secondary Objectives:** To assess long-term efficacy, vector persistence (transgene levels), and the incidence of new malignancies or autoimmune disorders.

2.2 Background & Rationale Per Health Authorities guidelines for gene therapy medicinal products that utilize integrating vectors (e.g., lentiviral vectors), long-term safety and efficacy follow-up of treated patients is required. Collecting data on the long-term effects of CAR-T cell therapy will help to further define the risk-benefit profile of CAR-T therapies and ensure patient safety over a 15-year horizon.

3. Study Design & Methodology

3.1 Design Framework Study Type: Observational / Long-Term Follow-Up **Control Type:** Single-arm **Blinding:** Open-label **Randomization:** N/A

3.2 Planned Enrollment & Duration Target \$N\$: 1,400 participants **Number of Sites:** Multicenter (Global sites including University of Pennsylvania and Novartis-supported locations) **Estimated Study Start:** 02-Nov-2015 **Estimated Primary Completion:** 22-Feb-2036

4. Subject Selection (Eligibility Criteria)

4.1 Inclusion Criteria 1. All patients who have received a CAR-T therapy and completed or discontinued early from a Novartis-

sponsored treatment protocol that utilized CAR-T cells or from any CAR-T trial sponsored by the University of Pennsylvania with which Novartis has a contractual agreement to co-develop the CAR technology. 2. Patients who have provided informed consent for the long term follow up study prior to their study participation.

4.2 Exclusion Criteria 1. There are no specific exclusion criteria for this study.

4.3 Demographic & General Eligibility * **Age Requirement:** 0 Years to 100 Years * **Gender/Sex:** All * **Healthy Volunteers Accepted:** No

5. Intervention & Treatment Plan

5.1 Investigational Regimen Dosage/Frequency: N/A (Patients are rolling over from a primary treatment protocol where they previously received their CAR-T infusion). **Route of Administration:** N/A for this follow-up phase (Primary therapy was intravenous). **Duration of Treatment:** 15 years of post-treatment follow-up from the last CAR-T treatment.

5.2 Concomitant & Prohibited Therapy Permitted: Standard of care for underlying conditions, including subsequent anti-cancer therapies if the primary disease progresses. **Prohibited:** No specific restrictions, as this is an observational safety follow-up.

6. Study Timeline & Visit Schedule

Visit ID	Window (Days)	Key Activities/Procedures	:---
:---	:---	Screening	N/A Informed Consent, Roll-over from primary CAR-T trial
Baseline	Day 0	Entry into LTFU study upon completion or early discontinuation of primary protocol	Interim
Years 1 to 14	Long-term safety monitoring, efficacy assessments, RCL testing, vector persistence labs	End of Study	Year 15 Final safety follow-up, Exit Interview

7. Outcome Measures (Endpoints)

7.1 Primary Endpoint Definition: Incidence of delayed adverse events (AEs) and detectable replication competent lentivirus (RCL) over the 15-year period. **Measurement Method:** Clinical assessments and centralized molecular/laboratory analysis for RCL.

7.2 Secondary & Exploratory Endpoints * Incidence of new malignancy. * New incidence of exacerbation of rheumatologic disorder, autoimmune disorder, or hematologic condition. * Vector persistence (transgene levels).

8. Safety & Adverse Event Reporting

Adverse Event (AE) Monitoring: Continuous long-term monitoring via routine clinic visits and laboratory assessments. **Serious Adverse Events (SAEs):** Reported per standard pharmacovigilance guidelines for gene therapy. **Data Monitoring Committee (DMC):** Internal and external safety monitoring committees managed by Novartis.

9. Statistical Analysis Plan (SAP) Summary

Analysis Populations: All enrolled patients exposed to CAR-T therapy. **Sample Size Justification:** Target $N = 1,400$ based on the anticipated number of patients treated across Novartis/University of Pennsylvania CAR-T clinical trials. **Handling of Missing Data:** Standard observational study imputation methods (e.g., Last Observation Carried Forward or multiple imputation) for longitudinal safety parameters.

10. Quality Assurance & Regulatory Compliance

GCP Compliance: This study will be conducted in accordance with Good Clinical Practice (GCP) and specific Health Authority guidelines for gene therapy medicinal products utilizing integrating vectors. **Monitoring Plan:** Periodic site monitoring and remote data capture across the 15-year duration. **Audit Trail:** Strict eCRF locking and tracking of long-term patient status to ensure compliance with gene therapy regulatory mandates.

11. Study Identifiers & Governance

Primary ID: CCTL019A2205B **Registry Number:** NCT02445222 **Sponsor/Lead Organization:** Novartis Pharmaceuticals **Study Title:** CAR-T Long Term Follow Up (LTFU) Study (PAV0) **Official Regulatory Title:** Long Term Follow-Up of Patients Exposed to Lentiviral-Based CAR T-Cell Therapy

12. Clinical Framework (CAR-T Specific)

Primary Condition: Post-treatment Follow-Up for Leukemia/Lymphoma (Previously treated CAR-T patients) **Diagnosis Threshold:** Patients previously exposed to CAR-T cell therapy **Medical Methodology:** Long-Term Observational Follow-Up for Gene Therapy **Optimization Target:** Identification of delayed AEs, RCL, and long-term vector persistence

13. Study Procedures & Methodology

Management Pathway: Standard long-term survivorship pathway for CAR-T recipients **Education Protocol (HCP):** Training on identifying and reporting delayed gene therapy toxicities and RCL **Education Protocol (Patient):** Informed consent detailing the 15-year commitment and signs of late AEs **Quality Audit Mechanism:** Centralized safety database tracking and routine queries

14. Observation & Follow-Up Schedule

Total Duration: 15 Years (Follow-up) **Onsite Visit Cadence:** Aligned with standard of care oncology survivorship visits **Remote Monitoring Frequency:** Periodic health status updates as determined by the protocol **Checklist Review Interval:** Annually (or per specific intervals dictated by the primary study protocol)

15. Sign-off & Approval

Role	Name	Signature	Date	---	---	---	---
Principal Investigator	[Insert Name]	_____	[DD-MMM-YYYY]				
Clinical Operations Lead	[Insert Name]	_____	[DD-MMM-YYYY]				
Lead Statistician	[Insert Name]	_____	[DD-MMM-YYYY]				